FORENSIC CASE STUDY: THE STALE FRAME TREE COLLISION

Autopsy of an Industrial Manipulator Joint Jam Caused by Asynchronous Coordinate Frame Desynchronization

1. CAMERA FRAME ARRIVAL (t = 0 ms)

- Vision pipeline estimates PCB socket target pose T_C^S
- Processed in user-space Linux background thread

2. WORKCELL VIBRATION (t = 35 ms)

- Mechanical fixture shifts 4.2 mm due to pneumatic ejector
- Frame tree is not updated because camera frame is still rendering

3. BLIND ACTION DISPATCH (t = 60 ms)

- Action planner emits trajectory chunk targeting stale pose
- Manipulator moves at 1.2 m/s with zero compliance

4. PHYSICAL JAM & OVER-TORQUE (t = 78 ms)

- Pin collides with fixture edge; motor current spikes to 18 A
- Gearbox teeth stripped; emergency E-stop tripped

FORENSIC ROOT CAUSE: Un-timestamped coordinate frame transforms without proprioceptive compliance.